Numerical Analysis of Conjugate Heat Transfer in a Double-Pipe Heat Exchanger

Thermal interaction between counter-flowing streams and a conducting separating wall

1. Objective

This experiment examines heat exchange between hot and cold laminar fluid streams flowing in separate passages with a solid wall between them. The study considers how the cold-side Reynolds number and the thermal conductivity ratio of the wall relative to the fluid affect the temperature field and outlet conditions.

2. Model and Simulation Settings

A two-dimensional conjugate heat-transfer model is considered. Convective and diffusive energy transport in both fluid regions is coupled with conduction through the separating wall. Parabolic velocity profiles are prescribed in the passages so that the analysis focuses on thermal transport.

Computational parameter	Value used
Streamwise grid points	160
Cold-fluid cross-stream points	20
Solid-wall cross-stream points	6
Hot-fluid cross-stream points	20
Cold-side Reynolds number	50, 100, 200
Hot-side Reynolds number	50, 100, 200
Solid-to-fluid heat-transfer ratio	1 and 100
Time step	1.25×10^3 (as stated)
Iterations	3000

3. Numerical Method

The energy equation is solved using a finite-volume convection–diffusion formulation in the hot and cold channels. The solid separating layer is modelled through the heat-conduction equation. The fluid and solid domains are coupled at their common interfaces, allowing heat to pass from one stream to the other.

4. Boundary Conditions

The inlet temperatures of the hot and cold fluids are specified. At the outlet boundaries, zero-gradient conditions are applied. The outer boundaries are assumed to be adiabatic. At each fluid–solid interface, both temperature and heat flux are maintained continuously to represent conjugate heat transfer.

5. Parameters and Outputs Considered

The simulations compare different Reynolds-number cases and two solid-to-fluid thermal ratios. The quantities examined include the hot- and cold-stream outlet temperatures, heat-transfer rate, effectiveness, temperature residual, relative energy imbalance and temperature contours.

6. Reported Results

The following table reproduces the outlet-temperature values listed in the supplied report. Temperatures are normalized, and the solid/fluid ratio is shown for each case.

Cold-side Re	Solid/fluid ratio	Hot outlet (normalized)	Cold outlet (normalized)
50	1	0.98	0.040
51	1	0.94	0.039
100	1	0.97	0.023
200	1	0.98	0.020
100	100	0.96	0.040
200	100	0.965	0.030

Note: The original results table lists a cold-side Reynolds number of 51 in its second row, while the parameter list specifies 50, 100 and 200. This value has been retained exactly as reported.

7. Discussion

The temperature-residual curves decrease as the iterative solution progresses, showing that the numerical temperature field moves toward convergence. The outlet-temperature histories also become nearly constant after the initial iterations, indicating that the outlet conditions have stabilized.

The temperature contours illustrate the thermal gradient across the two fluid passages and the separating solid region. The hot stream remains at a higher normalized temperature, while the cold stream gains heat through the wall. The extent of the temperature variation is affected by the Reynolds number and by the solid-to-fluid thermal ratio.

The later part of the supplied report also presents a detailed counter-flow case at Reynolds number 51 for both streams, with a Prandtl number of 7 and a conductivity ratio of 1. For this case, the reported solution reaches approximately 7,783 iterations and a final L-infinity residual of 1.0 × 10■■. The normalized outlet temperatures are 0.96054 for the hot stream and 0.03946 for the cold stream.

8. Conclusion

The experiment demonstrates the numerical modelling of conjugate heat transfer in a double-pipe heat exchanger. Heat is transferred from the hot passage to the cold passage by conduction through the separating wall, producing a decrease in hot-fluid outlet temperature and an increase in cold-fluid outlet temperature.

The residual and outlet-temperature plots provide evidence of iterative convergence, while the temperature contours show the distribution of thermal energy across the fluid and solid regions. The reported results indicate that Reynolds number and the relative thermal conductivity of the wall influence the computed thermal field.

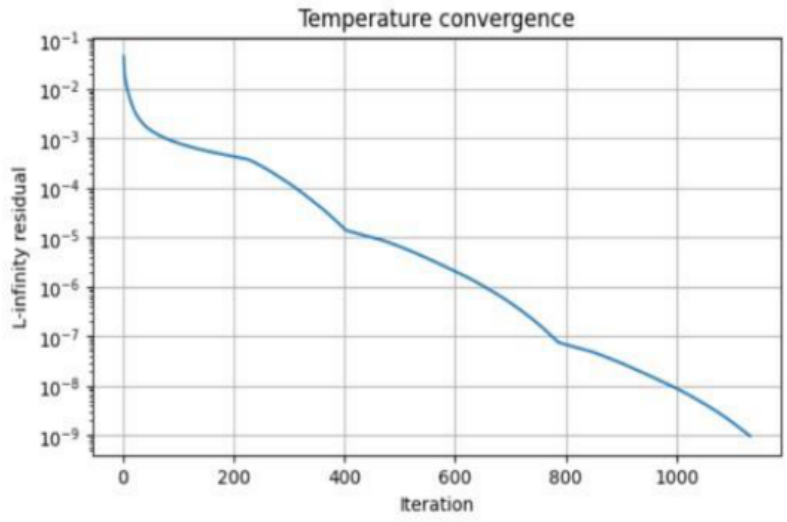
The present model uses a planar geometry and prescribed velocity profiles. Its results should therefore be interpreted within these assumptions. A more detailed cylindrical geometry, a coupled flow-field solution and comparison with experimental measurements could be considered in further work.

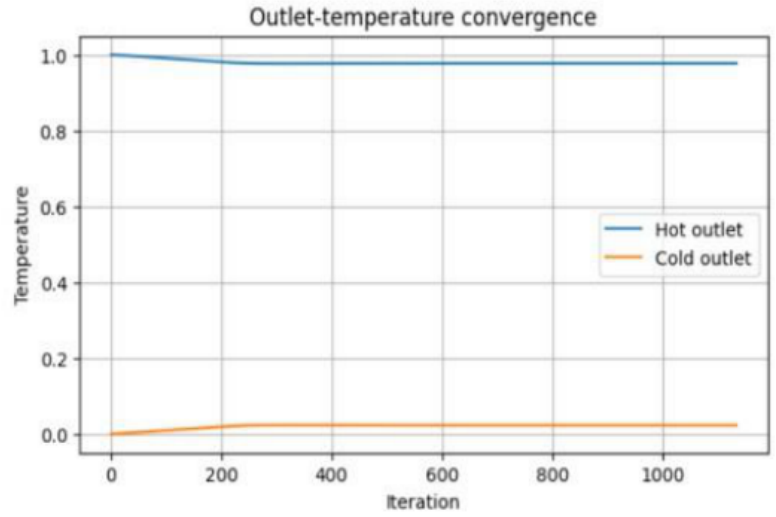
9. Simulation Figures

The original convergence histories and temperature-contour plots are retained in the following pages for reference.

Original simulation output (source page 2)

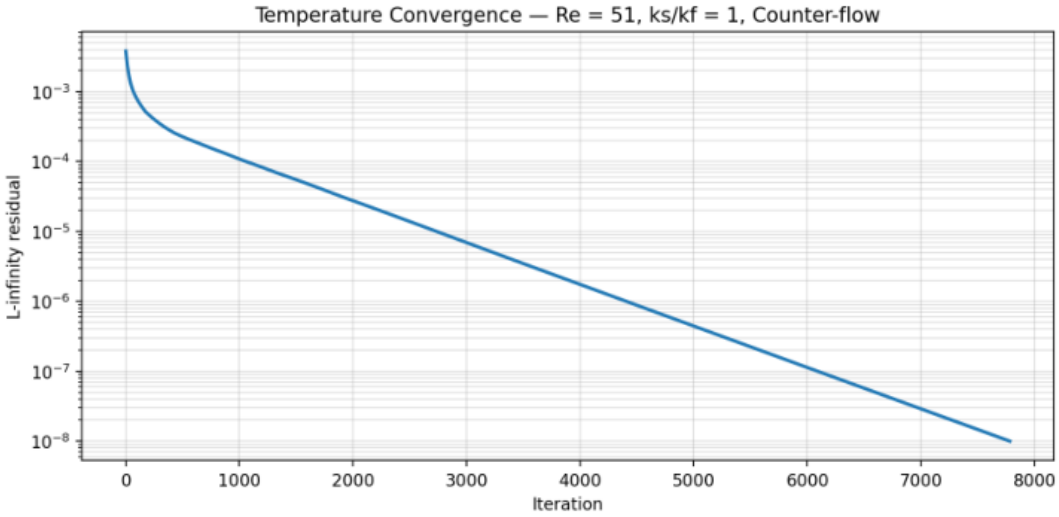
For Counter flow system Raynold's number 50 and wall to fluid ratio 1

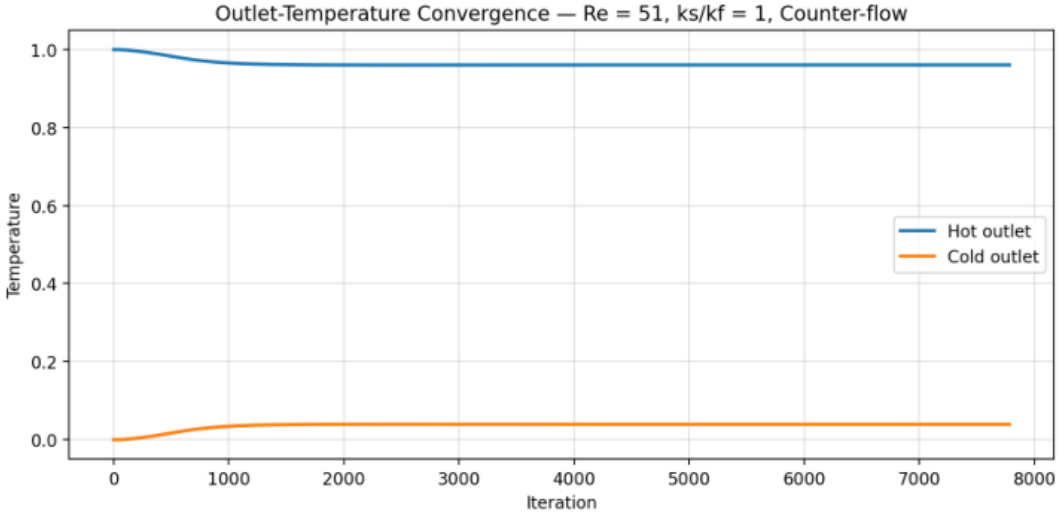




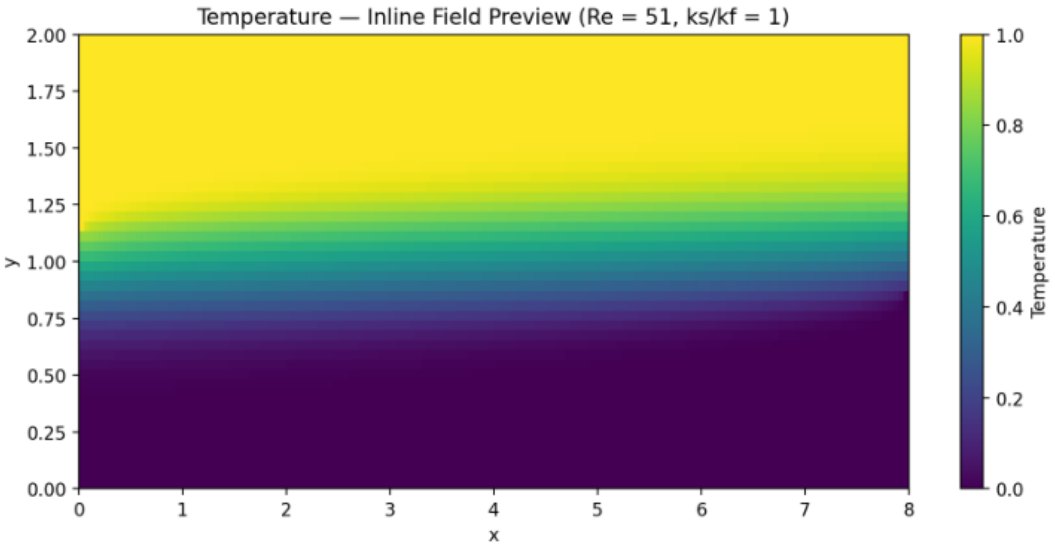
Original simulation output (source page 3)

RE =51 RESULTS

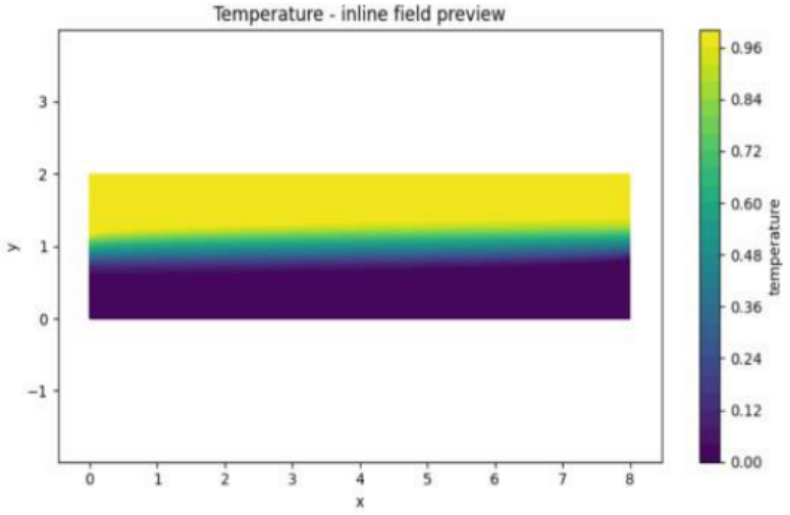




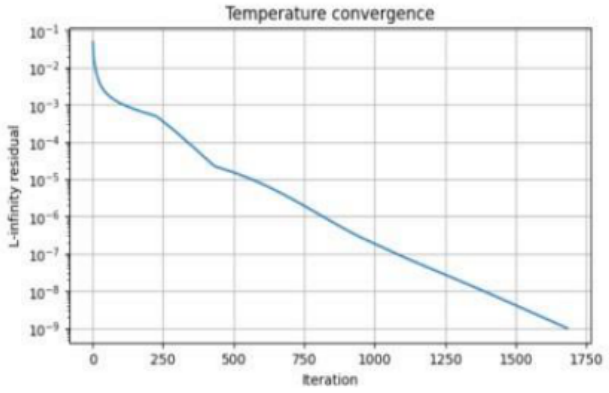
Original simulation output (source page 4)



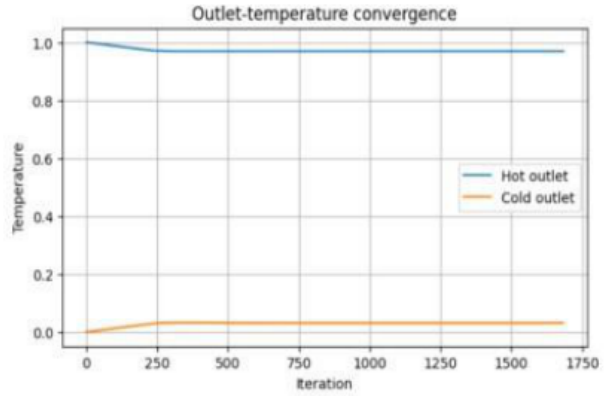
Original simulation output (source page 5)

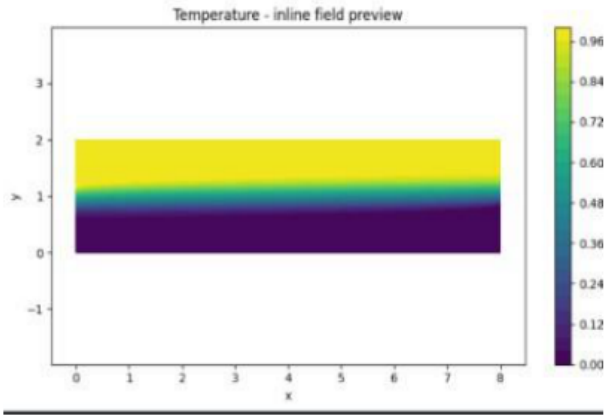


For Counter flow system Raynold's number 100 and wall to fluid ratio 1



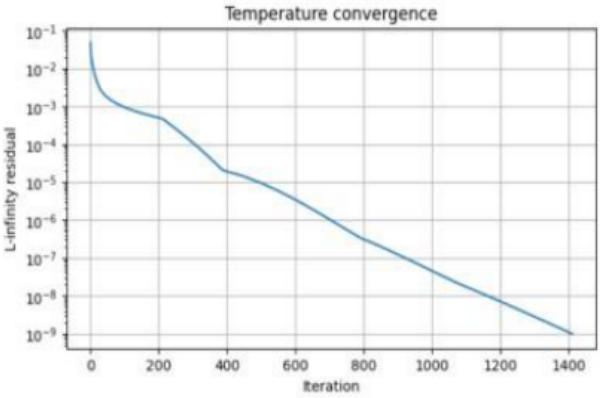
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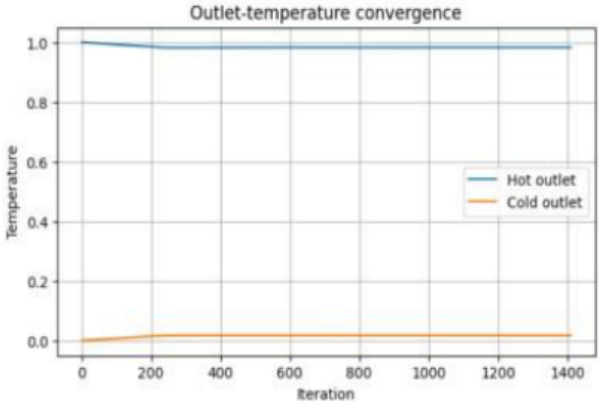




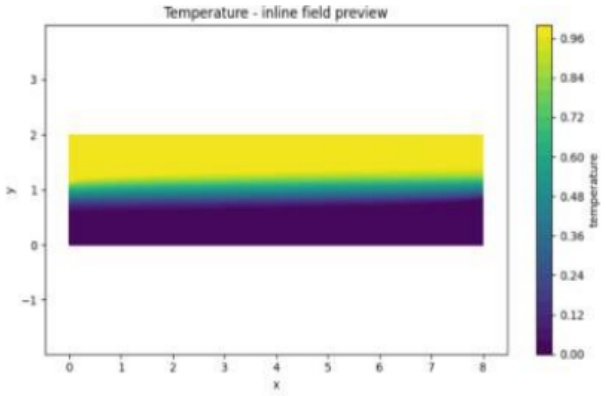
Original simulation output (source page 7)

For Counter flow system Raynold's number 200 and wall to fluid ratio 1

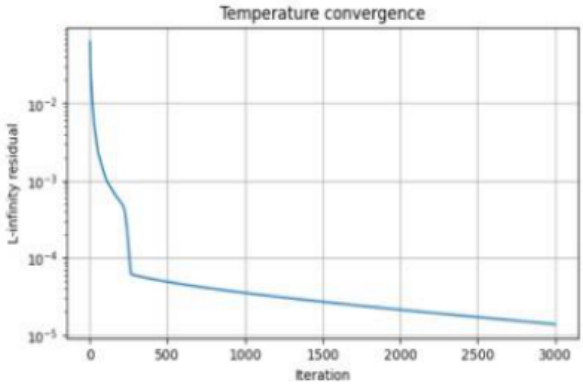




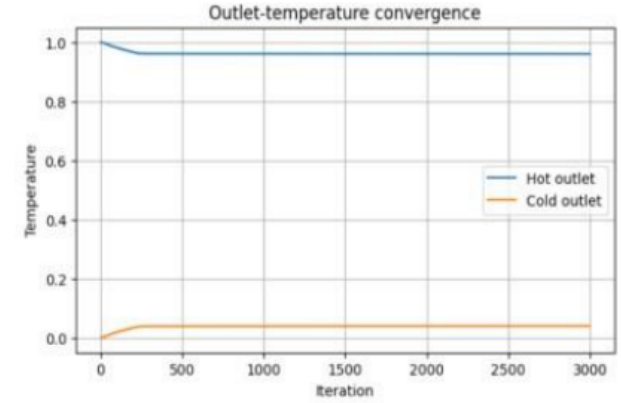
Original simulation output (source page 8)

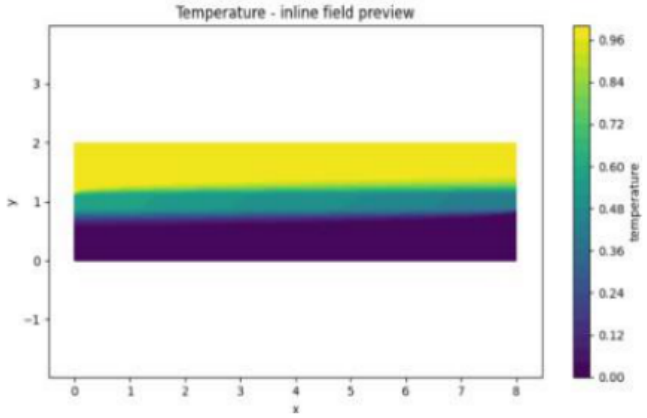


For Counter flow system Raynold's number 100 and wall to fluid ratio 100



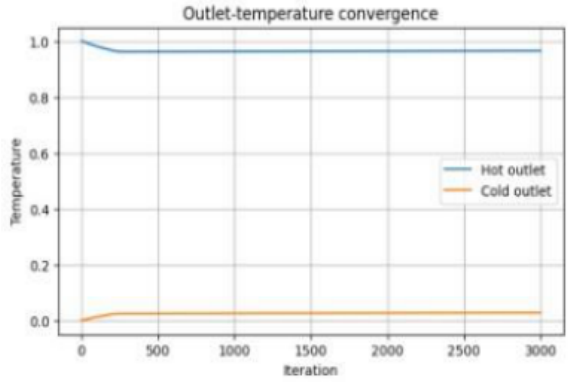
Original simulation output (source page 9)

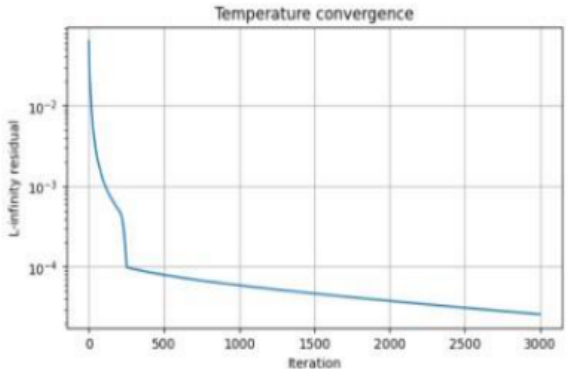




Original simulation output (source page 10)

For Counter flow system Raynold's number 200 and wall to fluid ratio 100





Original simulation output (source page 11)

Results and Discussion

A numerical study of conjugate heat transfer in a double-pipe heat exchanger was carried out using the Finite Difference Method (FDM). The simulation was performed under counter-flow conditions with Reynolds numbers of 51 for both the hot and cold streams. The Prandtl number was maintained at 7, and the wall-to-fluid thermal conductivity ratio was set to 1. The inlet temperatures were normalized to 1.0 for the hot fluid and 0.0 for the cold fluid.

The numerical solution converged after 7,783 iterations, reaching a final L-infinity residual of approximately 1.0×10^{-8}. This indicates that the temperature field approached the specified convergence criterion.

The normalized hot-fluid outlet temperature was 0.96054, while the normalized cold-fluid outlet temperature was 0.03946. The decrease in hot-fluid temperature and the increase in cold-fluid temperature indicate heat transfer from the hot stream to the cold stream through the separating solid wall.

The temperature field shows the distribution of temperature across the two fluid channels and the solid wall. The residual convergence graph demonstrates the reduction in numerical error during the iterative solution. The outlet-temperature convergence graph shows the stabilization of the hot and cold outlet temperatures as the solution progresses toward steady state.

These results demonstrate the application of the finite difference approach to modelling conjugate heat transfer in a simplified double-pipe heat exchanger configuration.

Conclusion

The conjugate heat transfer problem for a double-pipe heat exchanger was numerically investigated using the Finite Difference Method under counter-flow conditions. For Reynolds numbers of 51 in both streams and a wall-to-fluid thermal conductivity ratio of 1, the solution converged after 7,783 iterations with a final residual of approximately 1.0×10^{-8}.

The normalized outlet temperatures were 0.96054 for the hot stream and 0.03946 for the cold stream. These results confirm the expected direction of heat transfer from the hot fluid to the cold fluid through the separating wall.

The residual and outlet-temperature convergence plots provide a means of assessing the numerical stability and convergence of the solution. The temperature-field visualization further illustrates the spatial temperature distribution across the fluid and solid regions.

The model uses a planar representation of the double-pipe geometry and a prescribed velocity field. Therefore, the results should be interpreted within these modelling assumptions. Future work could include a more realistic cylindrical geometry, a more detailed velocity solution, and validation against experimental data.